

AYUSHI RAJ

AI-Powered Full Stack Software Engineer | Backend & Systems Specialist

Bihar, India | ayushi29507@gmail.com | ayushiraj.me | github.com/Silenttears-cloud | linkedin.com/in/alrya404

PROFESSIONAL SUMMARY

AI Product Engineer specializing in LLM orchestration, intelligent web applications, and full-stack API integration. Experienced in building multi-model API gateways, zero-knowledge cryptographic vaults, and autonomous Python execution sandboxes using TypeScript, Node.js, Fastify, and FastAPI.

TECHNICAL SKILLS

AI Engineering & Gateway: Multi-Model Routing • LLM Orchestration • Redis Rate Limiting • Prometheus Telemetry • Gemini API

Backend & Microservices: Node.js • Express.js • Fastify • Python (FastAPI) • C# (.NET Core) • RESTful APIs • WebSockets

Frontend UI & Full Stack: React.js • Next.js • Express.js • TypeScript • Redux • Monaco Editor • TailwindCSS

Databases & Caching: Redis • PostgreSQL • MongoDB • SQLite • Supabase • ChromaDB Vector Search

Languages: C • Rust • TypeScript • JavaScript • Python • C# • Java • SQL

Testing & DevOps: Jest • PyTest • xUnit • Docker • CI/CD • GCP • Git • Vercel • Render

TECHNICAL PROJECTS

Axiom AI Gateway - Low-Latency Multi-Model LLM Gateway (TypeScript, Node.js, Express, Redis, Docker)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Architected a resilient LLM orchestration gateway that routes AI requests across OpenAI, Anthropic, and Gemini endpoints to eliminate provider downtime.
- **Latency & Concurrency:** Designed an in-memory Redis token-bucket rate limiter and response cache, reducing downstream API latency by 45% under high request volume.
- **Reliability & Telemetry:** Integrated Prometheus telemetry metrics and structured JSON audit logging to maintain 99.9% gateway uptime with zero silent request drops.

Astra Vision - AI Sandbox & Code Parsing Platform (FastAPI, Python, Monaco, Tree-Sitter)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Architected an automated code-parsing platform and browser sandbox that lets developers safely execute untrusted code and visualize repository dependency graphs.
- **Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security filters to catch dangerous file system and network calls in real time.
- **Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph analysis under 200ms.

Alrya Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Engineered a client-side zero-knowledge password vault where user credentials are encrypted directly in the browser before reaching server storage.
- **Cryptographic Engineering:** Implemented Web Crypto API standards using AES-GCM (256-bit) payload encryption and PBKDF2 key derivation across 100,000 hashing rounds.
- **Performance & Security:** Developed high-throughput Express.js sync APIs with JWT authentication and strict CORS headers, keeping vault sync latency consistently under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Engineering Virtual Simulation

June 2026

- **Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET financial API controllers handling MongoDB document updates.
- **Action Taken:** Extended Web API controllers with `$set` atomic update operators and built an interactive React/Redux Goal Manager component for banking clients.
- **Engineering Outcome:** Wrote automated xUnit and Moq test suites covering complex edge cases to guarantee high uptime and reliable transactions.

Y Combinator (YC Startup - Shiptivity) - Software Engineering Virtual Simulation

June 2026

- **Problem & Scope:** Fixed key constraint collisions and slow re-renders on a drag-and-drop Kanban task board used by remote engineering teams.
- **Action Taken:** Designed atomic SQLite transaction logic for real-time task reordering and wired Dragula state hooks inside a React container.
- **Engineering Outcome:** Reduced UI render lag by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js build configurations.

Walmart USA - Advanced Software Engineering Virtual Simulation

May 2026

- **Problem & Scope:** Solved performance bottlenecks in high-volume retail inventory processing queues handling unnormalized spreadsheet imports.
- **Action Taken:** Developed a Java K-ary Max Heap utilizing fast bitwise shift operators (`<<` , `>>>`) and structured 3NF database models with Factory patterns.
- **Engineering Outcome:** Accelerated queue indexing calculations by 35% and created Python ETL scripts using `csv` and `sqlite3` to clean incoming inventory data.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida

Expected July 2028

- **Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)